

Technical Document #1046: Data Engineering - Comprehensive Overview

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Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Data governance establishes policies and procedures for managing data assets. It covers data privacy, security, and compliance.

Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Key Takeaways:

- Data lakes store raw data in its native format.
- Schema evolution handles changes in data structure over time.
- Data warehouses store structured data optimized for analytical queries.